HOSSAIN MAHMUD

Process Engineer | Chemical Engineer | Materials & Equipment Specialist
hussaindpi95@gmail.com | Sohar, Oman — Open to Relocate Worldwide
linkedin.com/in/hossain-mahmud-19320a146 | github.com/Hossain-Mahmud00/cantera-projects

PROFESSIONAL SUMMARY

Chemical Engineer with over 8 years of experience in process engineering, industrial production operations, chemical processing, and plant optimization within manufacturing and process industries. Skilled in process monitoring, production optimization, equipment troubleshooting, chemical operations, and quality control procedures. Possesses strong expertise in chemical process design, heat and mass transfer, fluid mechanics, reaction engineering, and industrial safety operations. Experienced in handling plant utilities, process equipment, production systems, and operational analysis while ensuring compliance with HSE standards, regulations, and operational procedures. Builds computational models in Python for reaction kinetics, reactor simulation, thermodynamics, vapor-liquid equilibrium, distillation column design, and heat exchanger sizing, with projects published on GitHub. Proficient in Aspen HYSYS, Python, Cantera, AutoCAD, MATLAB, ANSYS, and Microsoft Office.

CORE COMPETENCIES

Chemical Engineering: Process Engineering | Chemical Processing | Heat Transfer | Mass Transfer | Reaction Engineering

Process Operations: Production Optimization | Process Monitoring | Plant Operations | Utilities Management | Process Control

Industrial Systems: Fluid Mechanics | Distillation Systems | Utility | Chemical Equipment | Process Troubleshooting | Plant Reliability

Quality & Safety: Quality Control | Process Safety | HSE Compliance | Risk Assessment | Operational Safety Procedures

Process Analysis: Process Simulation | Process Improvement | Performance Monitoring | Operational Analysis | Optimization

Tools & Software: Aspen HYSYS | Python | Cantera | Thermo | NumPy | Matplotlib | AutoCAD | MATLAB | ANSYS | Microsoft Office

PROFESSIONAL EXPERIENCE

Materials Engineer — United Solar Polysilicon (FZC) SPC — Oct 2024 – Present

Sohar, Oman | 100,000 TPA Polysilicon Plant — TCS (Trichlorosilane) Manufacturing & Distillation

- **Commissioning & start-up:** Independently participated in the commissioning and start-up of a large-scale 100,000 TPA polysilicon TCS manufacturing and distillation plant, covering pre-commissioning, commissioning, and operational stabilization.
- Supported critical commissioning activities, including pipeline purging, target shooting, leak testing, line flushing, equipment inspection, and system integrity verification.
- Successfully started up 164 pumps, 102 heat exchangers, and 18 reboilers, including field verification, valve line-up, system checks, parameter monitoring, and troubleshooting.
- **Plant operations & monitoring:** Monitored and controlled production processes and equipment, including distillation columns, reactors, reboilers, heat exchangers, coolers, and pumps.
- Conducted systematic plant inspections to identify equipment abnormalities, process deviations, leaks, mechanical issues, and operational hazards, supporting safe and reliable operation.
- Performed quality control checks and adjusted process parameters as needed; maintained accurate logs of process conditions, incidents, maintenance, and shift handovers.
- **Safety & standards:** Implemented 6S management and ensured hygiene of machinery and equipment. Worked with multinational engineering teams, gaining practical experience in commissioning, plant operations, troubleshooting, and international industrial standards.
- Developed strong hands-on expertise in process safety, chemical plant HSE, process instrumentation, rotating equipment, process piping, chemical handling, troubleshooting, and field operations.

Senior Executive – Anodizing, ETP & Laboratory — KAI Aluminium — May 2023 – Oct 2024

Gazipur, Bangladesh

- Led anodizing, effluent treatment plant (ETP), and laboratory operations for a team of 20+, consistently meeting safety, quality, cost, and production KPIs.
- Drove continuous improvement initiatives, reducing process waste and improving throughput through systematic analysis and targeted operational changes.
- Managed departmental budgets, procurement approvals, and capital investment recommendations to enhance operational efficiency and equipment reliability.
- Ensured full regulatory compliance and proactively resolved customer quality concerns, maintaining a zero-major-defect record over 18 months.

Assistant Engineer – Utility — City Group **Aug 2020 – May 2023**

Narayanganj, Bangladesh

- Led end-to-end commissioning of water treatment systems, Chiller, Cooling Tower, AHU, FCU — pre-commissioning, live startup, and post-commissioning — ensuring full design compliance and handover.
- Optimized treatment processes (filtration, chemical dosing, disinfection, sludge handling) using SCADA control systems, reducing chemical consumption by optimizing dosing schedules.
- Coordinated cross-functional teams across operations, engineering, and construction to deliver process improvements on schedule and within budget.

Associate Engineer – Water Processing — Coca-Cola International Beverages Pvt. Ltd. **Feb 2017 – Jun 2020**

Mymensingh, Bangladesh

- Supervised installation and commissioning of WTP (Thermax) and Ultrafiltration System (Chirwa) to design specifications, maintaining compliance with international F&B standards.
- Maintained and calibrated measuring instruments; ensured full compliance with CCP, FSMS, HACCP, GMP, and BSTI standards in water processing.
- Investigated operating problems, performed root cause analysis, and implemented corrective actions, reducing unplanned downtime and improving OEE.
- Managed critical spare parts inventory and ensured production readiness after CIP (Clean-in-Place) and COP (Clean-out-of-Place) cycles.

COMPUTATIONAL PROJECTS (PYTHON)

- **Methane combustion (Cantera, GRI-Mech 3.0):** adiabatic flame temperature vs equivalence ratio, ignition delay vs temperature and pressure, and Arrhenius analysis of effective activation energy.
- **Plug flow reactor:** simulated ignition along a tube and quantified how mass flow rate sets the ignition distance.
- **Thermodynamics (Thermo library):** fluid properties, vapor pressure curves, and Peng-Robinson real-gas analysis showing about 20% density error from the ideal-gas assumption for methane at 200 bar.
- **Vapor-liquid equilibrium:** ethanol-water boiling diagram with Raoult's law and Van Laar activity coefficients; calculated azeotrope (x = 0.91, 351.45 K) compared with published data (about 0.89, 351.3 K).
- **TCS distillation column design (Thermo, Python):** trichlorosilane/silicon tetrachloride separation using Raoult's law and McCabe-Thiele; 22 theoretical stages at R = 1.83 (1.3 x Rmin) for illustrative feed and product specifications, cross-checked with Fenske-Gilliland (22.6 stages).
- **Heat exchanger sizing (TCS condenser):** first-pass chilled-water condenser design for a 183 kW duty; built overall U (797 W/m^2K) from Dittus-Boelter and Nusselt coefficients with wall and fouling resistances; 10.7 m^2, 30 tubes in 2 passes, 0.23 bar water-side pressure drop (example inputs).

CERTIFICATIONS & TRAINING

- Lead Auditor – ISO 45001:2018 (OHS Management)
- COSHH & Working in Confined Spaces
- Safe Handling of Process Plant Equipment — Alison, CPD Certified (1 September, 2026)
- OECD Good Laboratory Practices — UNIDO (United Nations), Online Course (3 October, 2026)
- Lean Six Sigma White Belt — GLSS, Cert. No. 0261000163 (3 October, 2026; valid to 2028)
- Executive Leadership — LinkedIn Learning (15 October, 2026)
- Operations Foundations: Total Productive Maintenance (TPM) — LinkedIn Learning (15 October, 2026)
- Production Management Foundations — LinkedIn Learning (15 October, 2026)

EDUCATION

Bachelor of Engineering (B.Eng.) – Chemical Engineering
Z. H. Sikder University of Science & Technology (ZHSUST), Bangladesh | 2017 – 2021
Evening program, completed while working full-time

Diploma in Engineering – Chemical Technology
Dhaka Polytechnic Institute, Bangladesh | 2012 – 2016

ADDITIONAL INFORMATION

Nationality: Bangladeshi | **Current Visa:** Valid Oman Work Visa | **Availability:** Open to relocate worldwide | **Languages:** Bengali (Native), Hindi (Fluent), English (Professional), Arabic (Basic) | **Industries Targeted:** Semiconductor, Oil & Gas, Water Treatment, Food & Beverage | **References:** Available on request